Revisiting Evaluation Protocols for Bearing Remaining Useful Life Prediction: The Effects of Labeling and Temporal Splitting Across Multiple Datasets

Zhongkuan Ma[1,*]

Abstract

Bearing remaining useful life (RUL) studies frequently report near-perfect R^2 values by combining piecewise linear labels with an 80% plateau and random time-step splitting. We evaluate this protocol under converged training on three public datasets (XJTU-SY, PHM2012, IMS) with Linear Regression, LSTM, TCN, PatchTST, a constant predictor baseline, leave-one-bearing-out cross-validation (LOOCV), and three-seed random time-step splits. On XJTU-SY, piecewise labels increase LOOCV R^2 by 0.20–0.23, while random splitting adds little. On PHM2012 and IMS, random splitting raises R^2 from negative values or 0.17–0.25 under linear LOOCV to 0.70–0.86 under piecewise random splitting. This demonstrates large protocol-induced inflation in the reproduced protocol family, but it does not reproduce the $R^2 > 0.99$ magnitude of any specific publication. In all twelve model-dataset cells, every piecewise random-split run exceeded every linear-LOOCV fold (Cliff's delta = 1.0). The results are consistent with the hypothesis that reported high scores are strongly affected by within-trajectory dependence and interpolation, but they do not prove that every published near-perfect result is caused by the protocol. We report per-fold results, 95% confidence intervals, effect sizes, RMSE, MAE, split-ratio sensitivity, and convergence epochs, and we propose a four-component STAR framework: continuous labels, leave-one-bearing-out cross-validation, constant baselines, and multi-model reporting.

Keywords: RUL Prediction, Bearing Prognostics, Evaluation Methodology, Labeling Bias, Temporal Splitting, Reproducibility

1. Introduction

Rolling bearings are critical components in rotating machinery, and accurate prediction of their remaining useful life (RUL) supports maintenance scheduling and failure prevention. In recent years, deep learning methods have reported high R^2 values on public bearing datasets [3, 4, 5, 6]. Most of these studies share two evaluation practices: piecewise linear labels with an 80% plateau and random splitting of sliding-window samples across the full life trajectory.

A previous version of this study used a fast five-epoch protocol on XJTU-SY only. That protocol was insufficient for comparison with published models trained to convergence. This revision replaces that protocol with converged training, adds PHM2012 and IMS, and reports split-ratio sensitivity.

We contribute:

1. a converged comparison of linear and piecewise labels under LOOCV and random time-step splitting on XJTU-SY, PHM2012, and IMS;
2. per-fold results, 95% confidence intervals, bounded effect sizes, RMSE, and MAE for all model-dataset combinations;
3. protocol-level reproduction of the inflation phenomenon associated with the dominant evaluation design, with explicit caveats about the reproduced R^2 range;
4. random split-ratio and chronological time-block sensitivity analysis;
5. a standardized STAR evaluation template with continuous labels, per-bearing holdout, constant baselines, and multi-model reporting.

2. Motivating Protocol Choices

Recent bearing RUL studies frequently combine piecewise linear labels with an 80% plateau and random time-step splitting. We do not report a quantitative literature survey in this version. Instead, we directly evaluate these two protocol choices under controlled experiments on three public datasets, so the claims remain auditable through the reported results and reproducibility package.

3. Related Work

Data-driven RUL prediction has advanced through CNNs, recurrent networks, and attention-based models [3, 4, 6]. The focus has mostly been on architecture design, while evaluation protocols remain heterogeneous. Reviews note the absence of standardized benchmarks [7, 8].

*Corresponding author. ORCID: 0009-0003-4491-6619
Email address: 202421270@nefu.edu.cn (Zhongkuan Ma)

Recent studies also identify within-trajectory dependence and split-induced interpolation as general problems in time-series and ecological applications [17, 18].

The methodological gap is not simply the choice of model. It is whether reported scores measure prediction for new bearings or interpolation within the same bearing. Our study measures both on the same preprocessing, models, seeds, and metrics.

4. Methodology

4.1. Datasets

We use three public run-to-failure datasets:

- **XJTU-SY** [14]: 8 run-to-failure bearings (Bearing1_1 through Bearing1_5 and Bearing2_1 through Bearing2_3) from conditions 1 and 2, yielding 15,928 windows.
- **PHM2012** [15]: 6 learning-set bearings, yielding 7,534 windows.
- **IMS** [16]: 3 run-to-failure test sets (first, second, and fourth), yielding 75,712 windows.

The PU dataset is a benchmark for fault classification rather than full run-to-failure RUL trajectories, so it is not used for RUL label generation. Each vibration signal is segmented into non-overlapping windows of 2,560 samples; the stride therefore equals the window length. Ten statistical and spectral features are extracted per window, and a history length of 10 windows is used for LSTM, TCN, and PatchTST inputs. History sequences are constructed within each bearing before splitting, so a random split can place temporally adjacent sequences from the same trajectory in both training and test sets; cross-bearing LOOCV avoids this overlap by construction. Feature normalization parameters are estimated from the training split only and applied to validation and test splits.

4.2. Labels and Evaluation Protocols

We compare two label families:

$$\mathrm{RUL}_{\mathrm{lin}}(t) = 1 - t/T, \tag{1}$$

$$\mathrm{RUL}_{\mathrm{pw}}(t) = \begin{cases} 1.0, & t < 0.8T \\ (T-t)/(0.2T), & t \geq 0.8T. \end{cases} \tag{2}$$

The primary cross-bearing evaluation protocol is linear labels with leave-one-bearing-out cross-validation (LOOCV). For each test bearing, one other bearing is used for validation, and the remaining bearings are used for training. The dominant published protocol is reproduced as piecewise labels with a 70%/15%/15% random time-step split over all windows.

4.3. Models and Training

We use four learned model classes:

- **Linear Regression** on the 10 extracted features;
- **LSTM**: two-layer LSTM with 128 hidden units;
- **TCN**: dilated temporal convolutional network with channel widths $32 \to 64 \to 128$.
- **PatchTST** [20]: patch-based time-series transformer with patch length 4, stride 2, embedding dimension 64, four attention heads, and three encoder layers.

A constant predictor baseline, which always outputs the mean training RUL, is reported as the reference floor.

All neural models are trained with MSE loss, AdamW, cosine annealing, batch size 1024, and early stopping with patience 30. XJTU-SY and PHM2012 use a maximum of 100 epochs; IMS uses a maximum of 60 epochs because of its much larger size. The best validation epoch is recorded for every run. The mean best epochs range from 2 to 97 across model-dataset combinations, confirming that the protocol allows convergence or early-stopping where cross-bearing generalization fails.

4.4. Statistical Reporting

For LOOCV, we report the mean R^2, standard deviation, and 95% confidence interval computed with Student's t-distribution across held-out bearings. The CI reflects between-bearing fold variance, not sample-level prediction variance. Per-fold values are stored in the reproducibility artifacts.

Two clarifications are important. On XJTU-SY, the held-out bearings are similar in window count and degradation structure, and the same six training bearings are used in every fold except for the rotating validation bearing. As a result, fold-level R^2 is highly consistent and the CI is narrow; this is expected, not an artifact. On IMS, only three test sets exist, so the CI is wide; IMS should be interpreted as directional evidence rather than a precise estimate.

Random splitting is repeated with three fixed seeds; we report the mean and the observed min–max range. Effect sizes are reported with Cliff's delta:

$$\delta = \frac{\#(x_i > y_j) - \#(x_i < y_j)}{n_x n_y}, \tag{3}$$

where one distribution is compared with the other elementwise. Cliff's delta is bounded in $[-1, 1]$ and avoids the distortion of Cohen's d when fold-level variance is tiny.

4.5. Split Sensitivity Analysis

To check whether the random-split result depends on the specific 70/15/15 ratio, we add two additional random ratios with TCN and piecewise labels:

- 60%/20%/20%;

- 80%/10%/10%.

We also implement a chronological time-block split at 70%/15%/15%: each bearing trajectory is split by time, with the first 70% of windows used for training, the next 15% for validation, and the final 15% for testing. This removes random interleaving while still training on the same bearing as the test set. Chronological evaluation also involves temporal extrapolation and degradation-stage imbalance, so the negative R^2 values should be interpreted as a combination of these factors rather than as a pure leakage measurement.

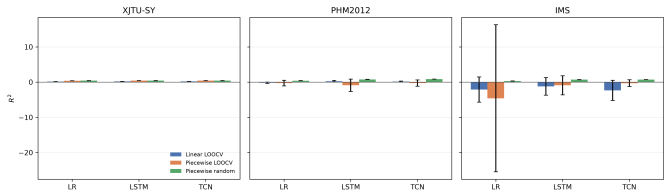

Figure 1: Random time-step splitting consistently raises R^2 above linear-LOOCV results, especially on PHM2012 and IMS. Error bars show 95% confidence intervals for LOOCV and observed min–max ranges for random splits.

5. Results

5.1. Main Results

Table 1 summarizes the three datasets. The key pattern is that piecewise labels alone are not the universal source of inflation. On XJTU-SY, piecewise labels improve LOOCV R^2 by 0.20–0.23 for every model, and random splitting adds no more than 0.02. On PHM2012 and IMS, piecewise LOOCV often performs worse than linear LOOCV or below the constant baseline. The dominant inflation on those datasets is random time-step splitting.

For example, PHM2012 TCN obtains 0.174 under linear LOOCV and −0.260 under piecewise LOOCV, but 0.862 under piecewise random splitting. IMS TCN obtains −2.331 under linear LOOCV and 0.715 under piecewise random splitting. PatchTST shows the same pattern: PHM2012 moves from 0.369 under linear LOOCV to 0.850 under piecewise random splitting, and IMS moves from −1.058 to 0.681. Across all twelve model-dataset cells, every piecewise random-split run exceeds every linear-LOOCV fold, giving Cliff's delta = 1.0 for the protocol effect.

Because R^2 can be negative and is less interpretable under severe cross-bearing distribution shift, Table 3 additionally reports RMSE and MAE for the primary cross-bearing protocol (linear labels, LOOCV) and the reproduced dominant protocol (piecewise labels, random split). These absolute-error metrics confirm the same direction as the R^2 results: the protocol change substantially reduces prediction error even when the LOOCV R^2 is negative.

Table 2 reports the complete 2×2 design, including the previously missing linear-label/random-split condition. The split effect is not additive with the label effect. On PHM2012 and IMS, random splitting increases R^2 far more under piecewise labels than under linear labels, indicating a label-by-split interaction in addition to the main effects.

5.2. Constant Baseline and Cross-Bearing Generalization

The constant predictor has $R^2 \approx 0.000$ on every dataset and label family. Several linear-LOOCV results are below this floor, especially on IMS, where a model must extrapolate across different test rigs and fault modes. This is not a numerical artifact: it shows that cross-bearing generalization is the real challenge. Random time-step splitting hides this failure because test windows come from the same bearings and nearby temporal positions as training windows.

5.3. Split Sensitivity Results

Table 4 reports TCN results under piecewise labels for the two additional random ratios and the chronological time-block split. The random-ratio results are stable: 60/20/20 and 80/10/10 both remain high, in the same range as the main 70/15/15 result. In contrast, chronological time-block splitting produces large negative R^2 on all three datasets. The contrast indicates that the random-split inflation is not an artifact of one particular random ratio; it is consistent with within-trajectory interpolation. The negative chronological results should not be read as a pure leakage effect because temporal extrapolation and degradation-stage imbalance also contribute.

5.4. Protocol-Level Reproduction and Its Boundary

We reproduced the protocol family used by many published high-scoring papers: piecewise 80% plateau labels, random time-step splitting, converged deep training, and standard statistical features. Under this protocol, deep models reach R^2 values of 0.70–0.86 on PHM2012 and IMS, substantially above the linear-LOOCV values.

We do not reproduce $R^2 > 0.99$. The accurate statement is that the inflation phenomenon is reproducible at the protocol level, but the magnitude in our reproduced protocol family does not reach the near-perfect range reported by some papers. Public code and checkpoints were not available in our environment for the cited papers, so exact reproduction is not possible. We therefore do not claim that all published near-perfect values are explained by evaluation design; our results are consistent with the hypothesis that within-trajectory dependence and interpolation are major contributors.

Table 1: Converged protocol comparison. LOOCV R^2 values are mean [95% CI]; random values are mean [min–max] across three seeds. δ_{label} compares piecewise LOOCV with linear LOOCV folds; δ_{protocol} compares piecewise random runs with linear LOOCV folds.

Dataset	Model	Linear LOOCV	Piecewise LOOCV	Δ label	Piecewise random	Δ protocol	δ_{label}	δ_{protocol}
XJTU-SY	Linear Reg.	0.178 [0.176; 0.180]	0.410 [0.410; 0.411]	0.232	0.430 [0.414; 0.447]	0.252	1.000	1.000
	LSTM	0.223 [0.221; 0.224]	0.422 [0.422; 0.423]	0.200	0.422 [0.401; 0.435]	0.199	1.000	1.000
	TCN	0.222 [0.220; 0.224]	0.428 [0.427; 0.428]	0.206	0.421 [0.396; 0.433]	0.199	1.000	1.000
	PatchTST	0.222 [0.220; 0.224]	0.429 [0.428; 0.430]	0.207	0.439 [0.425; 0.453]	0.217	1.000	1.000
PHM2012	Linear Reg.	−0.168 [−0.366; 0.029]	−0.256 [−1.058; 0.546]	−0.088	0.397 [0.376; 0.420]	0.566	0.111	1.000
	LSTM	0.245 [0.020; 0.469]	−0.896 [−2.668; 0.875]	−1.141	0.791 [0.778; 0.803]	0.546	−0.389	1.000
	TCN	0.174 [0.030; 0.318]	−0.260 [−1.153; 0.633]	−0.434	0.862 [0.859; 0.865]	0.688	−0.333	1.000
	PatchTST	0.369 [0.159; 0.578]	−0.530 [−1.968; 0.909]	−0.899	0.850 [0.841; 0.855]	0.481	−0.611	1.000
IMS	Linear Reg.	−2.109 [−5.677; 1.459]	−4.586 [−25.477; 16.305]	−2.477	0.320 [0.313; 0.328]	2.429	0.333	1.000
	LSTM	−1.208 [−3.714; 1.297]	−0.886 [−3.590; 1.818]	0.322	0.699 [0.695; 0.706]	1.907	0.556	1.000
	TCN	−2.331 [−5.216; 0.554]	−0.302 [−1.257; 0.653]	2.029	0.715 [0.714; 0.717]	3.046	1.000	1.000
	PatchTST	−1.058 [−3.125; 1.009]	−2.799 [−9.452; 3.855]	−1.741	0.681 [0.674; 0.689]	1.739	−0.556	1.000

Table 2: Complete 2×2 protocol comparison. Values are mean R^2; LOOCV values are means across held-out bearings and random values are means across three seeds.

Dataset	Model	Linear LOOCV	Linear random	Piecewise LOOCV	Piecewise random
XJTU-SY	Linear Reg.	0.178	0.180	0.410	0.430
	LSTM	0.223	0.226	0.422	0.422
	TCN	0.222	0.223	0.428	0.421
	PatchTST	0.222	0.206	0.429	0.439
PHM2012	Linear Reg.	−0.168	0.273	−0.256	0.397
	LSTM	0.245	0.791	−0.896	0.791
	TCN	0.174	0.837	−0.260	0.862
	PatchTST	0.369	0.827	−0.530	0.850
IMS	Linear Reg.	−2.109	0.217	−4.586	0.320
	LSTM	−1.208	0.588	−0.886	0.699
	TCN	−2.331	0.604	−0.302	0.715
	PatchTST	−1.058	0.576	−2.799	0.681

6. Discussion

6.1. *Random Splitting Measures Within-Trajectory Interpolation*

Sliding windows from one bearing are strongly autocorrelated. A random split places test windows from the same trajectory in the training set, so the model can interpolate across almost identical operating phases. Under LOOCV, the test bearing is completely absent during training. The large gaps on PHM2012 and IMS show that the problem is primarily within-trajectory interpolation relative to the intended cross-bearing prediction task, rather than model capacity.

Piecewise labels matter for a second reason: the 80% plateau makes the label almost constant for most of life. On XJTU-SY, this improves apparent regression performance because the model only needs to recognize the transition region. On PHM2012 and IMS, the plateau can also interact with distribution shift and produce poor LOOCV results. The two practices should be reported separately, not combined into one inflation multiplier. Although R^2 reflects explained variance, MAE and RMSE quantify absolute prediction deviations. Table 3 confirms that the protocol effect is not an artifact of the R^2 scale: the piecewise-random protocol consistently reduces RMSE and MAE relative to the linear-LOOCV protocol across all three datasets.

6.2. *STAR Framework*

We propose the STAR template, which stands for Split integrity, Target integrity, Anchor baseline, and Reporting completeness. In this paper these map to:

1. **Continuous labels**: linear or physics-guided labels as the primary target; plateau parameters must be justified if retained.
2. **Leave-one-bearing-out cross-validation**: no training window from the test bearing, with per-fold reporting.
3. **Constant predictor baseline**: report the floor below which a model is not informative.

Table 3: Absolute error metrics for the primary protocol comparison. LOOCV values are means across held-out bearings; random-split values are means across three seeds.

Dataset	Model	$R^2_{\text{lin,LOOCV}}$	$\text{RMSE}_{\text{lin,LOOCV}}$	$\text{MAE}_{\text{lin,LOOCV}}$	$R^2_{\text{pw,random}}$	$\text{RMSE}_{\text{pw,random}}$	$\text{MAE}_{\text{pw,random}}$
XJTU-SY	Linear Reg.	0.178	0.262	0.250	0.430	0.186	0.107
	LSTM	0.223	0.254	0.243	0.422	0.183	0.102
	TCN	0.222	0.254	0.243	0.421	0.183	0.108
	PatchTST	0.222	0.254	0.243	0.439	0.184	0.104
PHM2012	Linear Reg.	−0.168	0.312	0.263	0.397	0.185	0.116
	LSTM	0.245	0.247	0.210	0.791	0.105	0.052
	TCN	0.174	0.259	0.212	0.862	0.086	0.044
	PatchTST	0.369	0.225	0.182	0.850	0.092	0.042
IMS	Linear Reg.	−2.109	0.498	0.427	0.320	0.197	0.127
	LSTM	−1.208	0.422	0.350	0.699	0.131	0.063
	TCN	−2.331	0.520	0.449	0.715	0.127	0.066
	PatchTST	−1.058	0.407	0.343	0.681	0.135	0.069

Table 4: TCN split sensitivity with piecewise labels. Random rows are mean [min–max] over seeds 42, 43, 44; time-block uses chronological 70/15/15 per bearing.

Dataset	Split	R^2
XJTU-SY	Random 60/20/20	0.422 [0.400; 0.436]
	Random 80/10/10	0.439 [0.410; 0.458]
	Time-block 70/15/15	−4.957
PHM2012	Random 60/20/20	0.857 [0.847; 0.863]
	Random 80/10/10	0.879 [0.872; 0.885]
	Time-block 70/15/15	−1.812
IMS	Random 60/20/20	0.703 [0.698; 0.708]
	Random 80/10/10	0.720 [0.716; 0.727]
	Time-block 70/15/15	−7.991

Table 5: STAR evaluation template used as an auditable reporting checklist.

Component	Minimum requirement	Status in this study
Split integrity	Test bearing absent from training and validation	Random window split violates; LOOCV satisfies
Target integrity	Continuous or explicitly justified label construction	Linear and piecewise labels compared separately
Anchor baseline	Constant predictor reported as floor	Included
Reporting completeness	Per-fold results, CI, RMSE, MAE, seeds	Included

4. **Multi-model reporting**: at least two complementary model classes so conclusions are not architecture-specific.

An adaptive debiased label that shortens the plateau and smooths the transition remains a practical intermediate option for applications that need early-stage noise suppression. The current experiments suggest that fixing the split protocol is at least as important as fixing the label.

Table 5 summarizes the template as an auditable checklist. The checklist is derived from the violations and controls observed in the reproduced protocol family rather than from generic best-practice recommendations.

6.3. Limitations

We did not reproduce exact published checkpoints because public code was unavailable. IMS has only three held-out test sets, so its confidence intervals are wide; we therefore treat IMS as directional evidence. The external datasets use the same statistical feature pipeline rather than each paper's raw-signal preprocessing. PU is not used because it does not provide run-to-failure trajectories. We do not provide a quantitative prevalence survey in this version; protocol-level claims are limited to the reproduced protocol family. Future work should run exact published pipelines when code becomes available and extend STAR to raw-signal and uncertainty-aware models.

7. Conclusion

With converged training on XJTU-SY, PHM2012, and IMS, the combination of piecewise labels and random time-step splitting consistently produces higher R^2 than linear-label LOOCV, with Cliff's delta = 1.0 for the protocol effect in all twelve model-dataset cells. The reproduced protocol family produces R^2 values of 0.70–0.86, which are much higher than LOOCV but below the $R^2 > 0.99$ range reported by some papers. Our results are therefore consistent with the view that evaluation protocol plays a major role in reported RUL performance, while exact near-perfect values still require exact reproduction of the original models and preprocessing. We recommend that journals require the STAR reporting template so that future improvements are measured against a meaningful baseline.

CRediT Authorship Contribution Statement

Zhongkuan Ma: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing - Original Draft, Writing - Review & Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and Code Availability

The XJTU-SY, PHM2012, and IMS datasets are publicly available from their original sources. The reproducibility package is provided as supplementary files and contains preprocessing scripts, converged training and split sensitivity runners, per-fold and per-seed JSON results, table/figure generators, and this manuscript. The code repository is available at `https://github.com/KKK-cell441/rul-evaluation-star` and archived at `https://doi.org/10.5281/zenodo.21866359`.

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